

AIRCRAFT PROPULSION



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1 MONTH INTERNSHIP

ABSTRACT

The given report features the design of 3 models of ramjet and turbofan engines as per a problem statement and within certain parameters. A simulation and animation of the same has also been carried out. The necessary calculations will also be seen in the report. Brief information about the types of engines designed and the thought process and methodology will also be provided.

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INTRODUCTION

This internship has covered all types of powerplants used by aircraft such as reciprocating engines, turbofan engines, turbojet engines, scramjet engines, ramjet engines and the fundamentals of mathematics and physics needed for the final design. The project being discussed is a ramjet engine design based on the parameters given by calculating a problem statement. The problem statement is as follows:

$$\dot{m}_a = 80 \text{ Kg/s}$$

$$\dot{m}_f = 0.2 \text{ Kg/s}$$

$$V_e = 2.9 \text{ M}$$

$$V_i = 2 \text{ M}$$

$$P_0 = 1 \text{ bar (1 bar = 101325 Pa)}$$

$$P_e = 2.5 \text{ bar}$$

$$T = 70000 \text{ N}$$

METHODOLOGY

To solve the problem statement, it is a very simple matter of substituting the values into the following equations:

MASS FLOW RATE EQUATION:

$$\dot{m}_a = \rho \times A_i \times V_i$$

where,

- $\dot{m}_a$ = Mass flow rate of air in Kg/s
- ρ = Density of air (1.2 Kg/m³)
- V_i = Velocity at inlet in m/s
- A_i = Area at the inlet in m²

THRUST EQUATION:

$$T = (\dot{m}_a + \dot{m}_f)V_e - \dot{m}_a V_i + (P_e - P_0) A_e$$

where,

- T = Thrust in Newton
- $\dot{m}_a$ = Mass flow rate of air in Kg/s
- $\dot{m}_f$ = Mass flow rate of fuel in Kg/s
- V_e = Velocity at exit in m/s
- V_i = Velocity at inlet in m/s
- P_e = Pressure at exit in Pascal
- P_0 = Ambient Pressure in Pascal
- A_e = Area at exit in m²

The calculations for the project can be seen below:

Let A_e be x . Using thrust equation,

$$70000 = (80+0.2)994.7 - 80*686 + (253312.5 - 101325) X$$

$$70000 = (80.2)994.7 - 54880 + 151987.5X$$

$$124880 = 79774.94 + 151987.5X$$

$$45105.06 = 151987.5X$$

$$\underline{X = 0.296768 \text{ m}^2}$$

$$\underline{R_{Ae} = 0.30747 \quad D = 0.61494} \quad (\text{Using Area of circle formula})$$

From this first equation we get the Radius and diameter of the area of exit nozzle as well as its area.

Using Mass Flow Rate formula,

$$80 = 1.2 * A_i * 686$$

$$80 = 823.2 A_i$$

$$\underline{A_i = 0.097 \text{ m}^2}$$

Assuming area of front to be 0.282 (Trial and error method + relative sizing),

Area of nozzle = Area of front – Area of Inlet

$$0.282 - 0.097 = 0.185 \text{ m}^2$$

$$\text{Diameter of nozzle} = 0.485 \text{ m}$$

$$\underline{\text{Diameter of Inlet Inner chamber} = 0.60 \text{ m}}$$

$$\underline{\text{Diameter of Inlet outer chamber} = 0.733 \text{ m}}$$

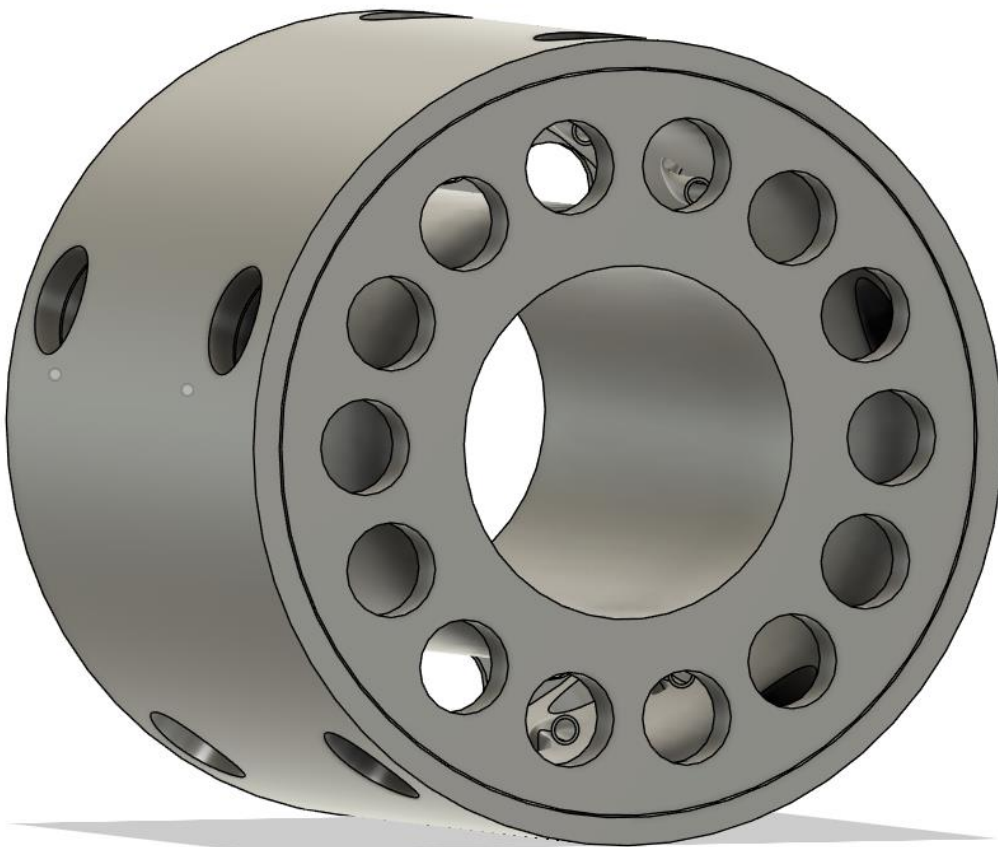
From this second equation we can get the area of inlet nozzle and thus the internal and external diameters.

The next step in the project would be the actual design of the 3d model of the ramjet engine. It is here that we switch the AutoCAD program Fusion 360.

In fusion 360 after using the 2d sketch option to make a basic shape from which to start (circular for the tube), the model was extruded and shelled to make the basic external section. After which the nozzles, shafts and later combustion chambers were added.

Two versions of the combustion chamber were made, namely an annular and can type. The models were made of titanium due to the high melting point which would be required given the heat given off by both the combustion and the skin friction due to the supersonic speeds at which the plane would be flying. The engine has been designed with a view of one sheet as an exterior hull which is why no space has been given in between the plates to account for the expansion of the metal as the sheet would expand as a whole. Along with these 2 models, 2 turbofan models were also created showcasing the intake fan, main shaft, low- and high-pressure compressors and turbines and 2 types of combustion chambers. The turbofan model was also animated to spin and also has a core and bypass chamber displayed. Images of all the models created can be seen below:

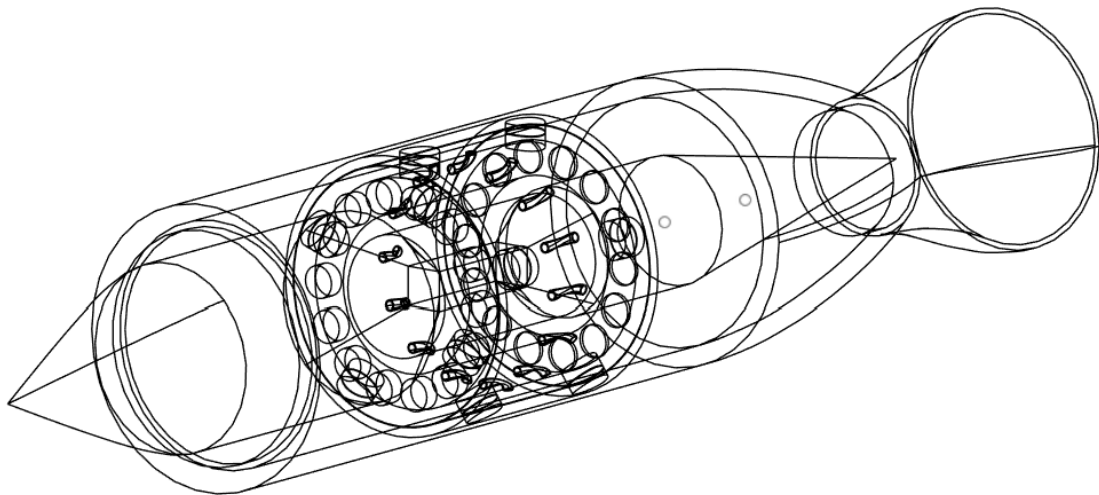
****Annular type combustion chamber featuring fuel nozzles, internal and external chambers, holes for air, dilution holes and a singular mixed chamber.****



****Central shaft of the ramjet engine with nozzles and the annular type combustion chamber****



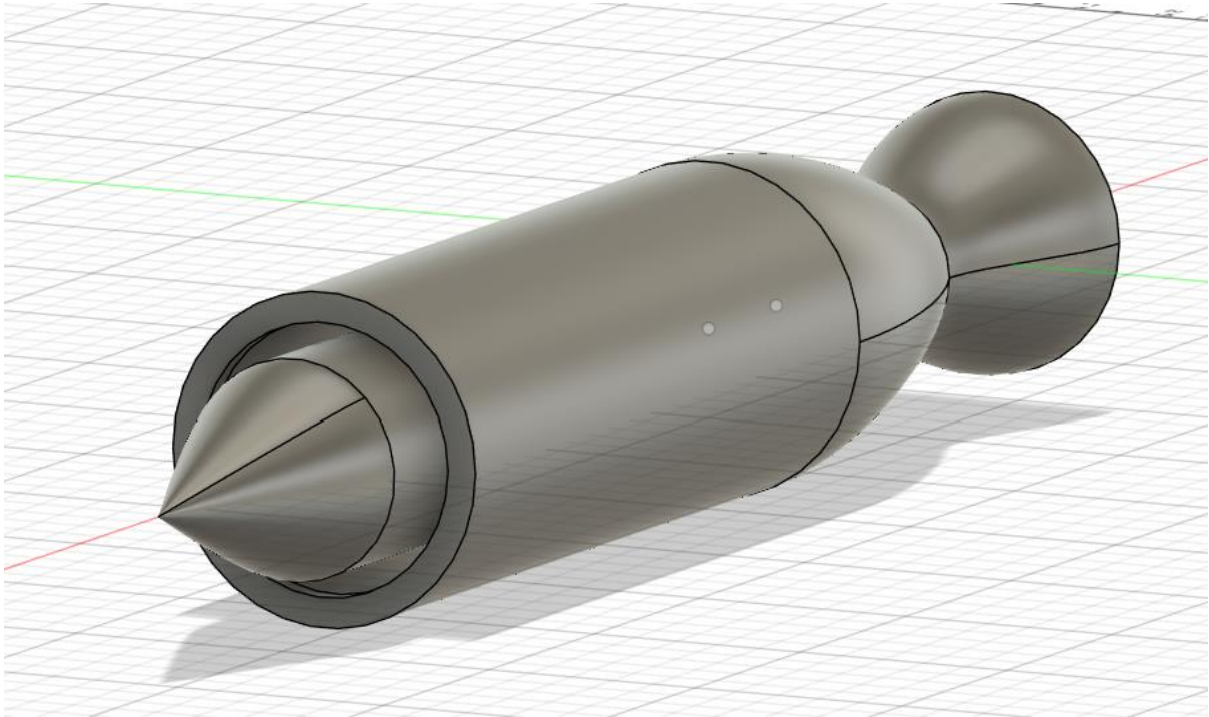
**** Wireframe view of the annular type ramjet engine****



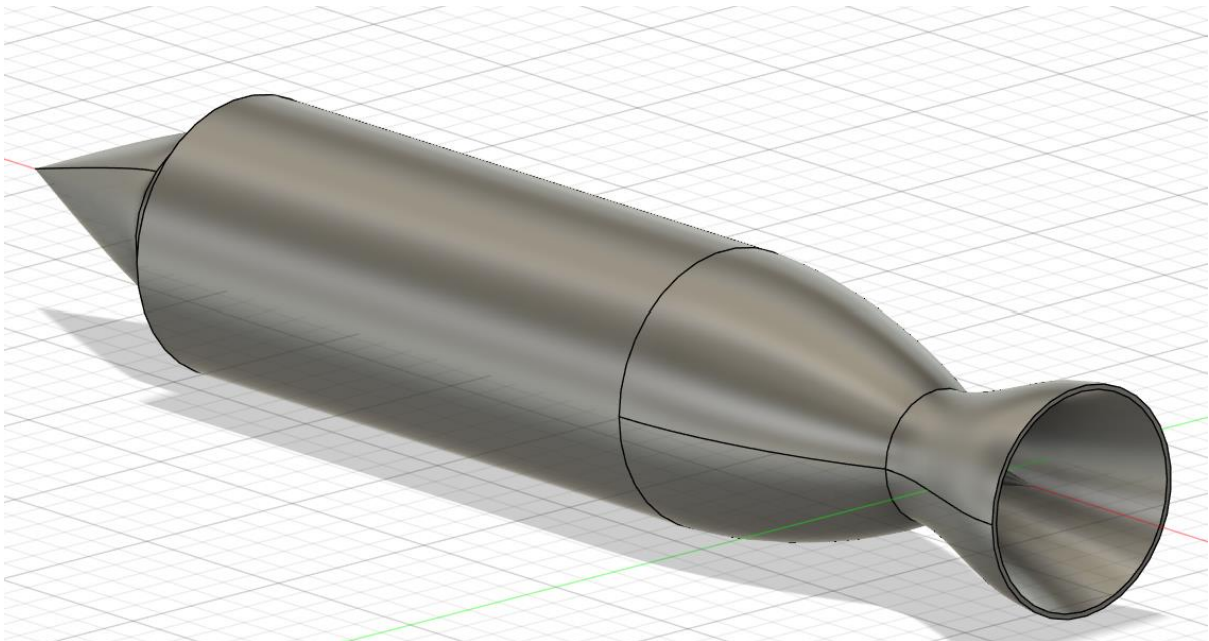
**** Front view of annular type ramjet engine****



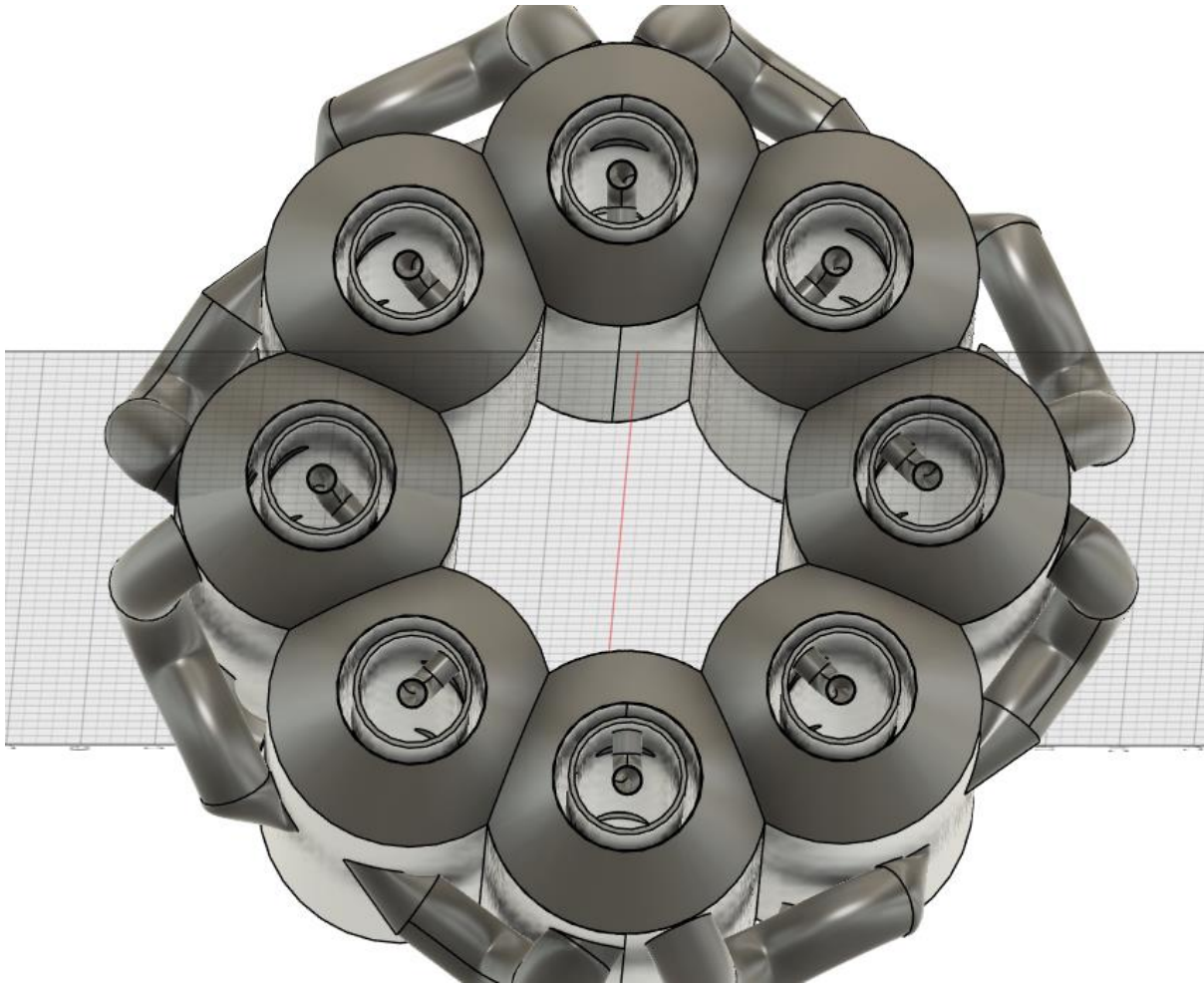
****Side profile of ramjet engine****



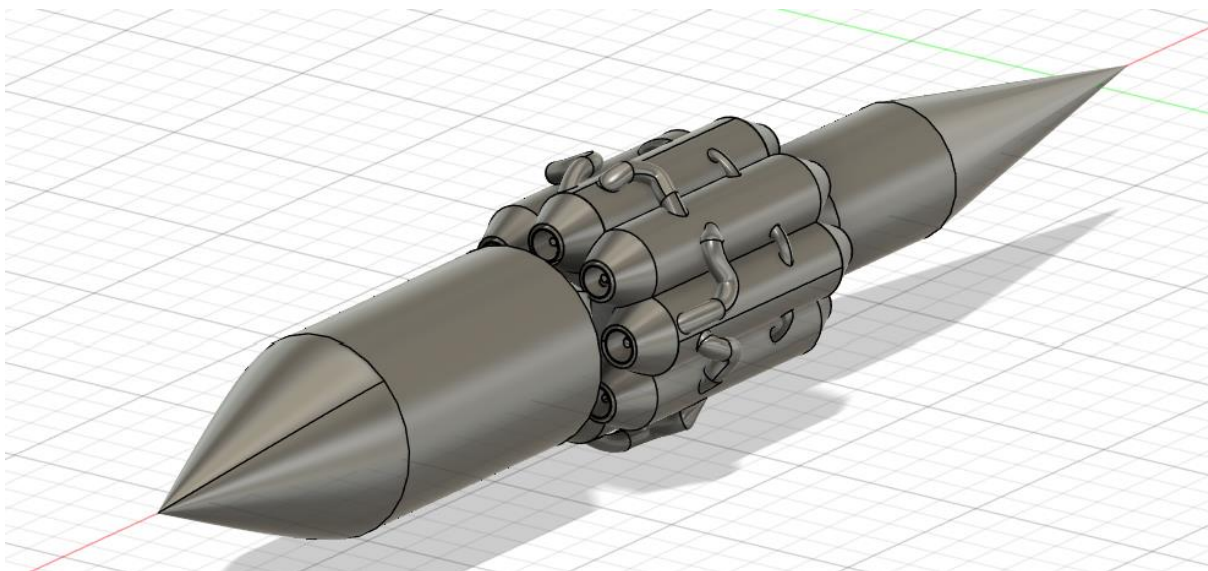
****Side back view of ramjet engine****



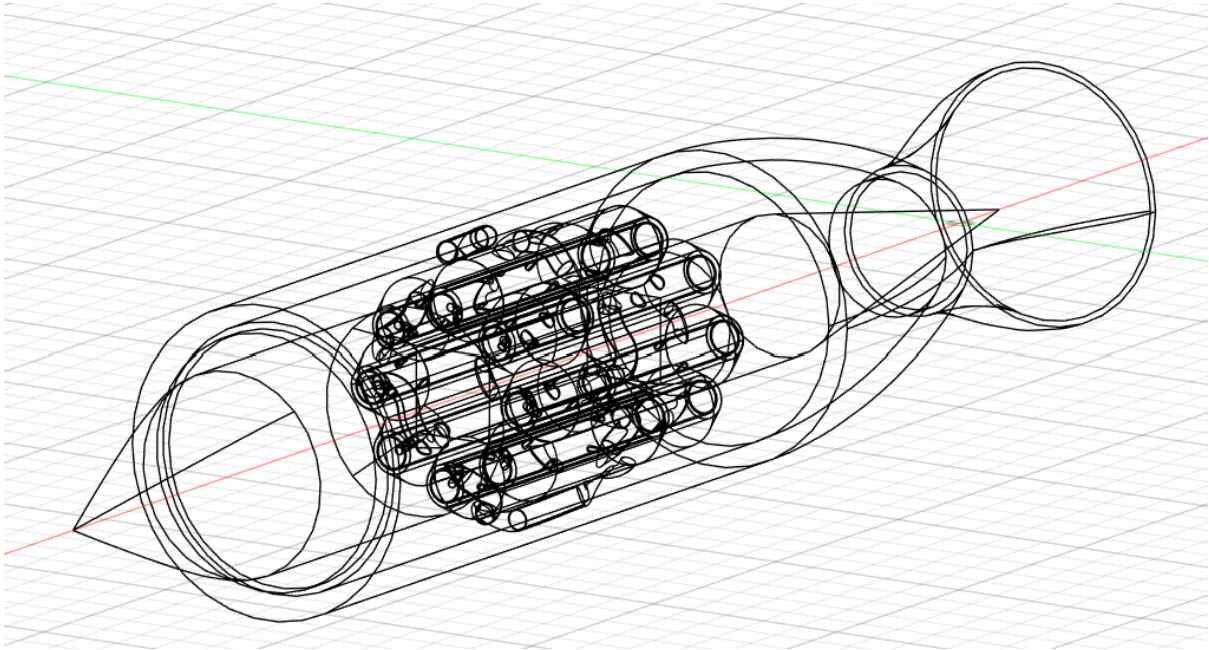
****Can type combustion chamber showcasing fuel manifolds, interconnecting pipes, fuel nozzles internal and external chambers, dilution holes and fuel nozzles****



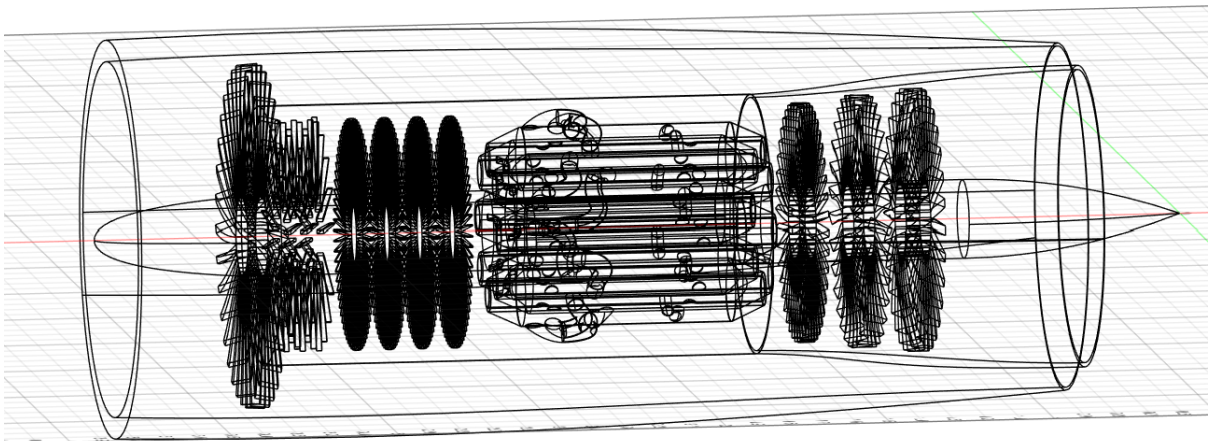
****Central shaft, nozzles and combustion chamber of can type ramjet engine****



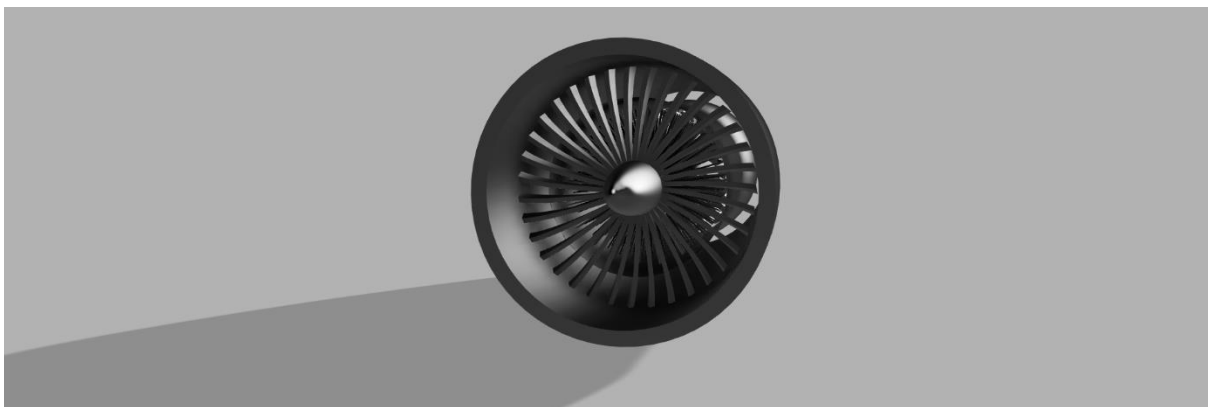
****Wireframe view of can type combustion chamber****



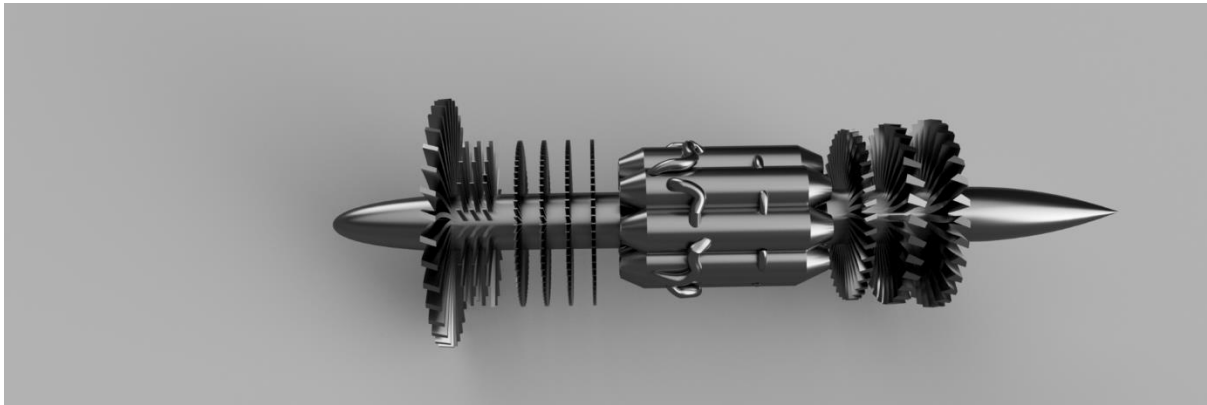
****Wireframe view of can type turbofan engine****



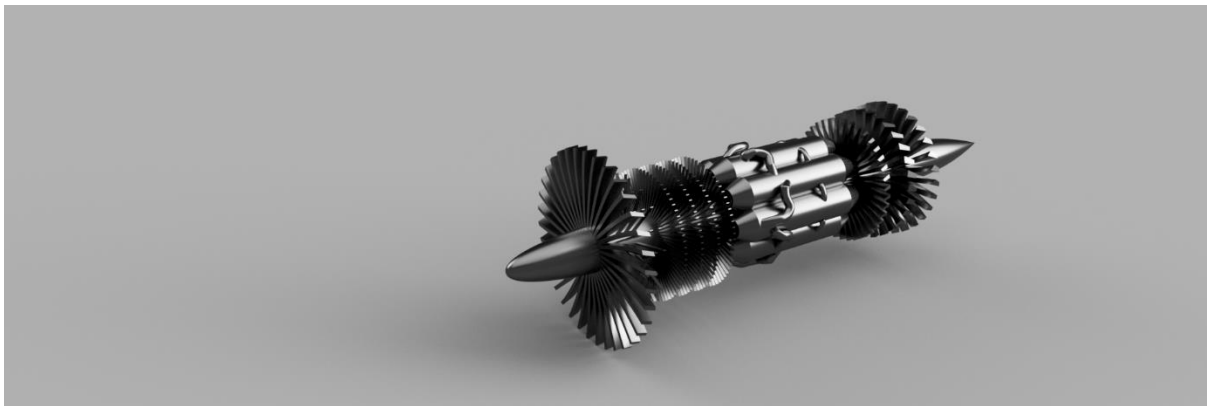
****Front profile of turbofan engine****



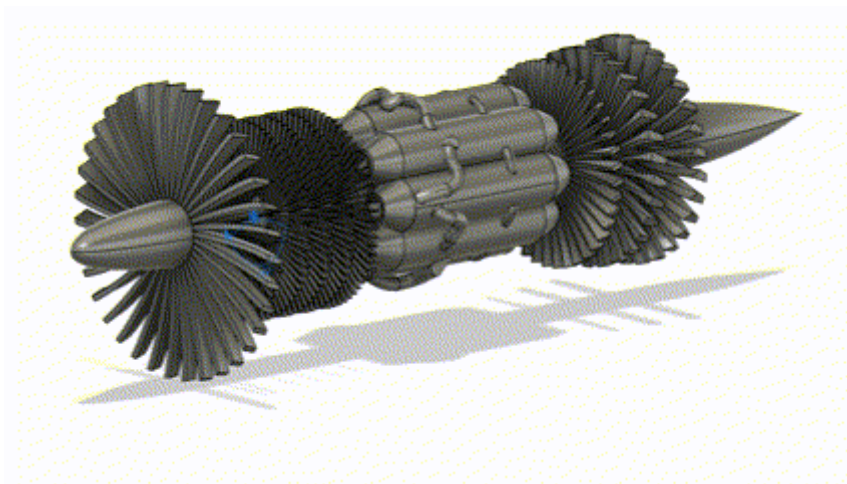
****Overhead view of can type turbofan engine internals****



****Side view of internals of can type turbofan engine****



****GIF showcasing animation of turbofan engine****



The dilution holes allow the air to mix, ensuring the pressure difference is not excessive, the fuel injectors provide the fuel for combustion. The blades of the turbofan stages were designed such that they would each excel at their individual tasks (short and relatively straight for compressors, turbines with a swirl to ensure sufficient rotation and a tilt angle for

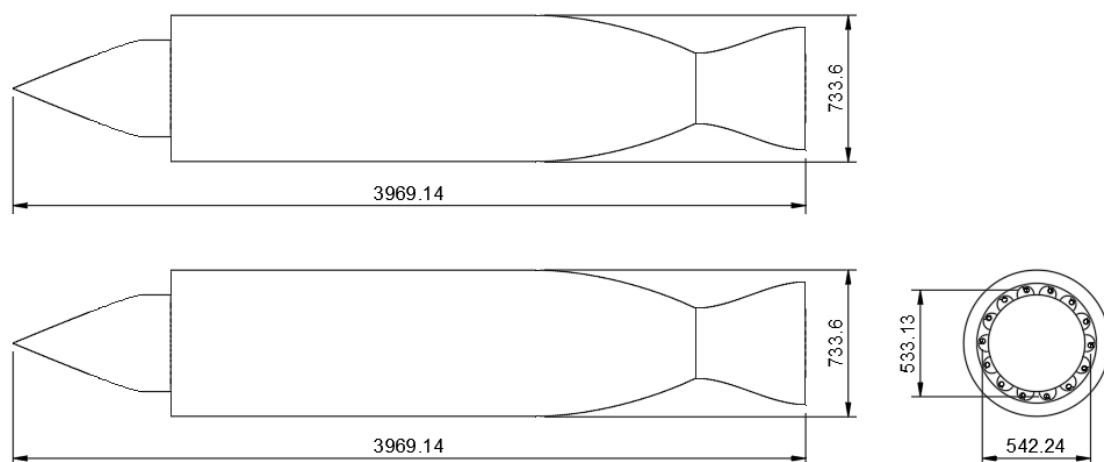
the front fan to ensure the air is pushed into the chamber while protecting the front from intake of excessive f.o.d. As seen:

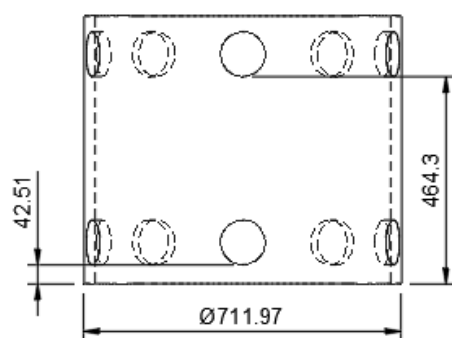
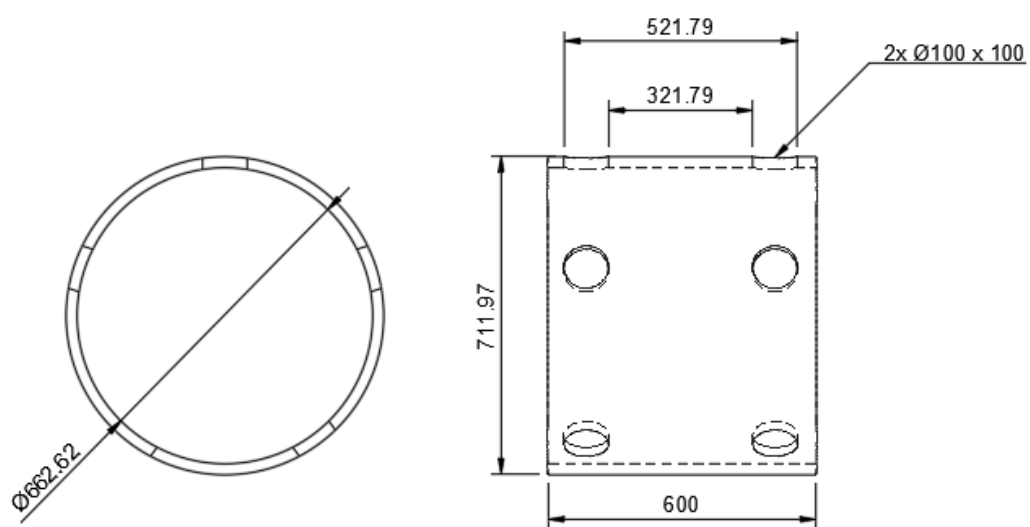
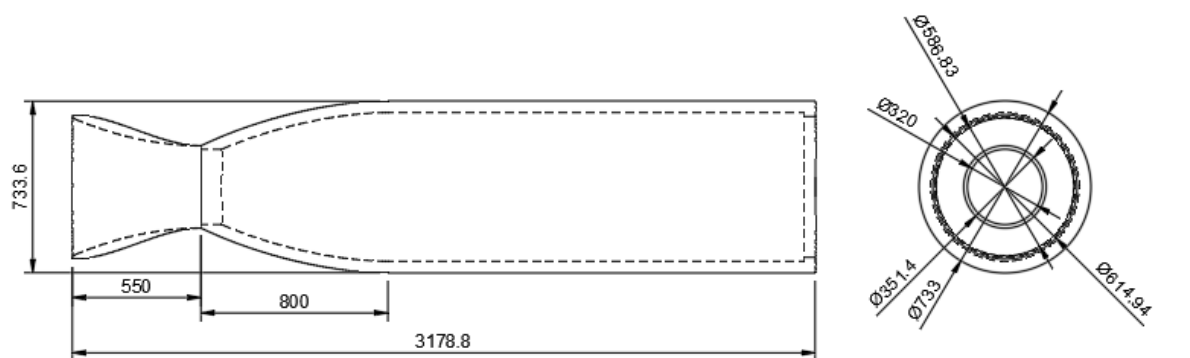
****Compressor: Turbine: Front intake blades****

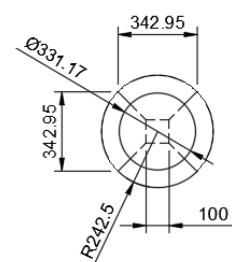
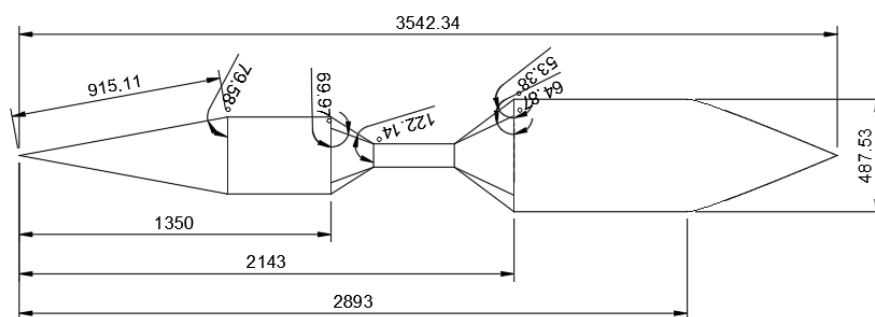
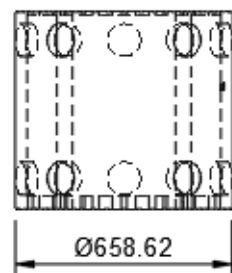
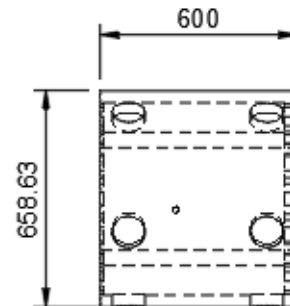
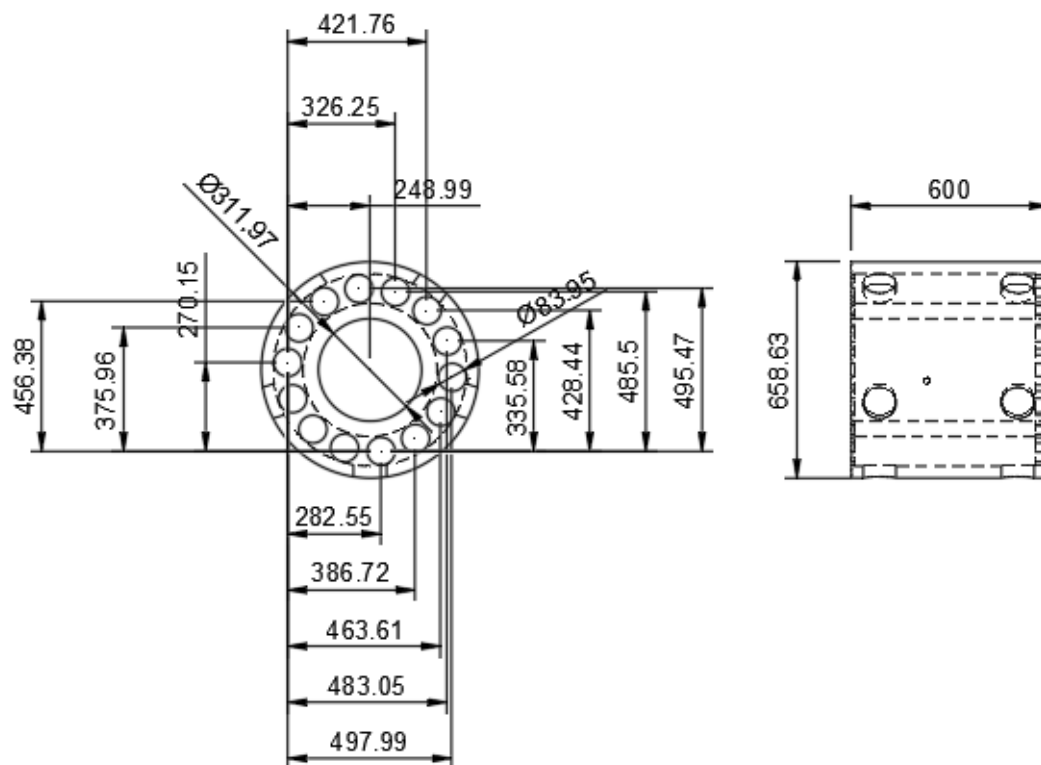


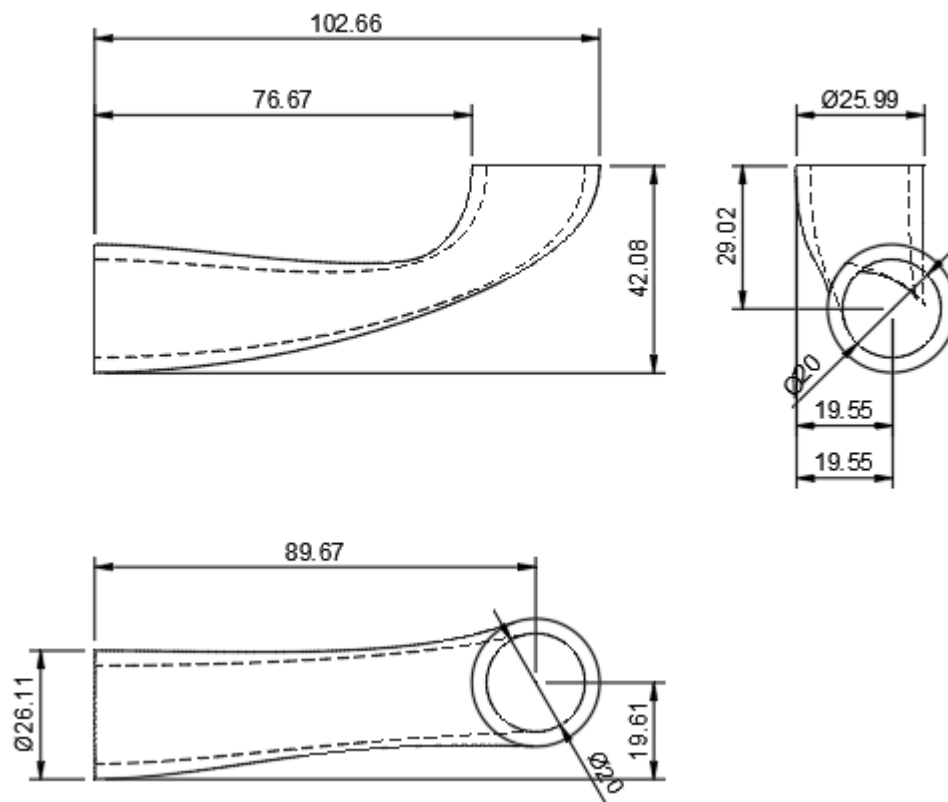
The nozzles of the turbofan are more curved than the ramjet as the ramjet needs a sharper edge to cut through the air at supersonic speeds while the turbofan will operate at subsonic speeds. The turbofan engine has an outer cowl and an inner separating wall. This creates a chamber for both the bypass air and core air. The bypass air in this engine is not especially high and the engine would be considered a low to medium bypass ratio engine.

A drawing of the ramjet engine was also created which specifies the exact dimensions and the design to be followed.







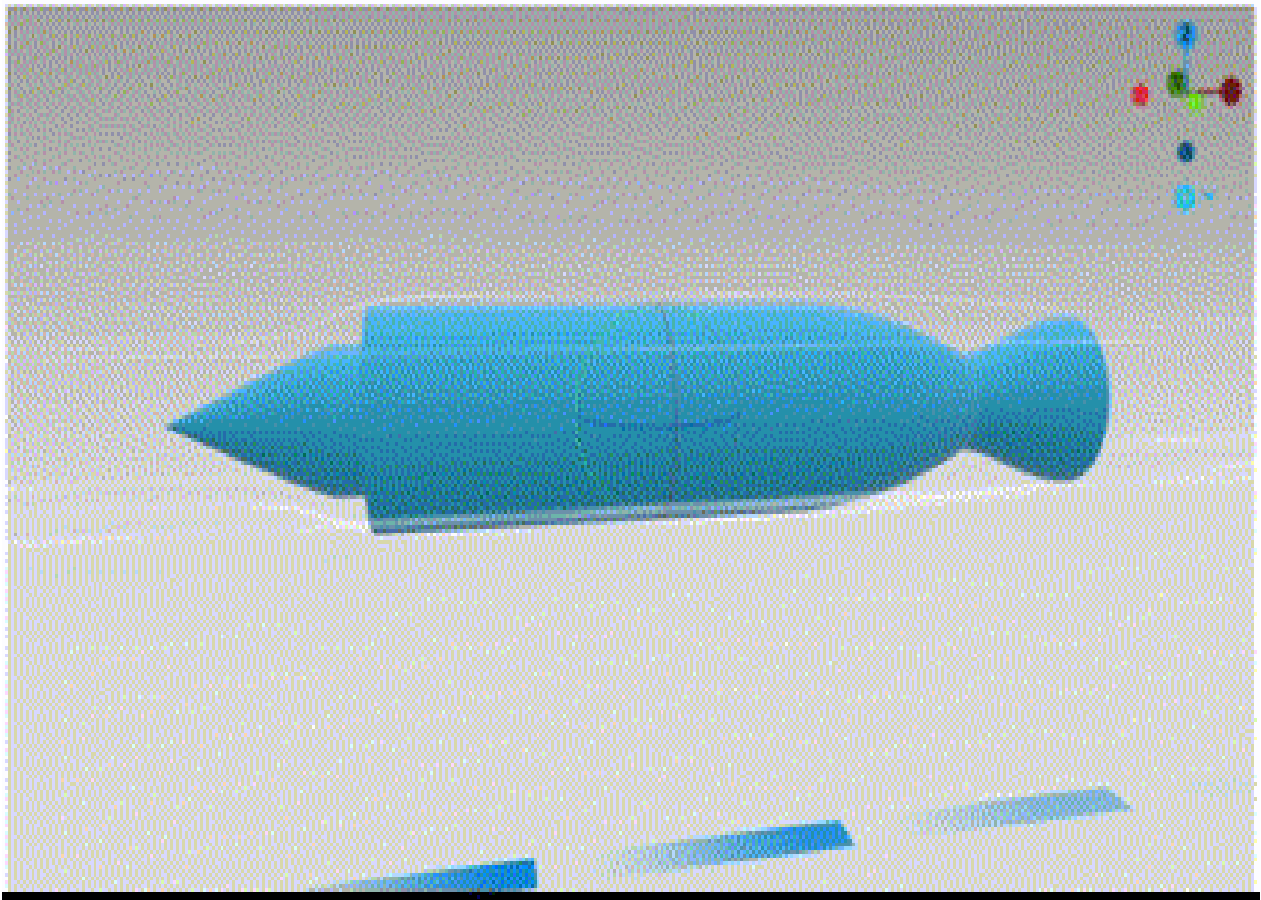


For the turbofan engines the 3d model was designed with aluminum due to its flexibility and strength. However, the design has the material Al-Li (Aluminum Lithium alloy) in mind which is not available in fusion. Al-Li is a new gen cheaper alternative to carbon composites.

SIMULATION AND ANALYSIS

The first simulation run is a wind tunnel test of the ramjet and turbofan engines; however, the internal airflow is not displayed as such software require payment. For the purpose of this report just the overall aerodynamics will be displayed. The result of the wind tunnel tests can be seen below (Speed of Mach 4 is assumed):

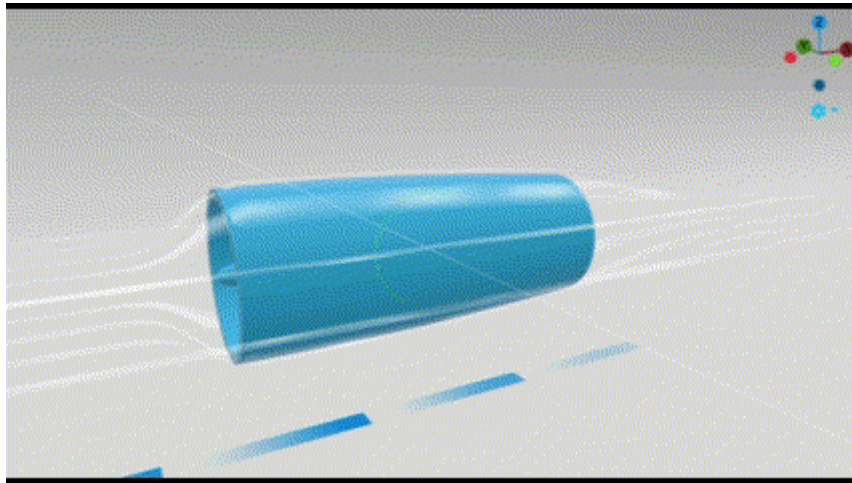
****Wind tunnel analysis of ramjet engine****



A smooth laminar flow is seen with exception of the bottom meeting point at end nozzle where a slight turbulent flow is seen which is not large enough to create an issue (very minimal drag increase).

Similarly for the turbofan engine (Speed of Mach 0.6 assumed):

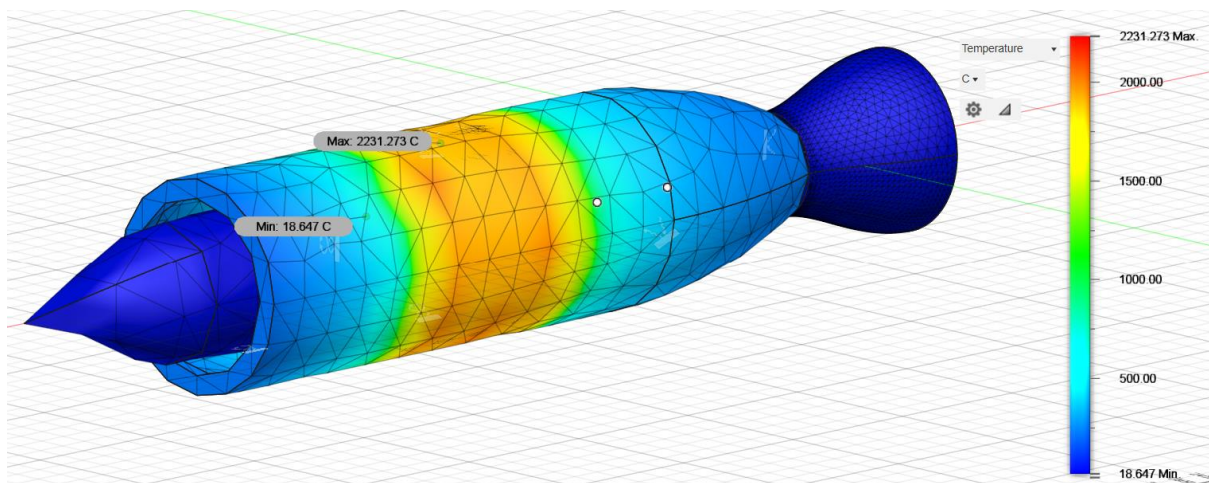
****Wind tunnel analysis of a turbofan engine****



A laminar airflow is seen for the turbofan engine with slight turbulence over the main cowling which is to be expected and will cause vortices only in extreme conditions of high angle of attack or moisture

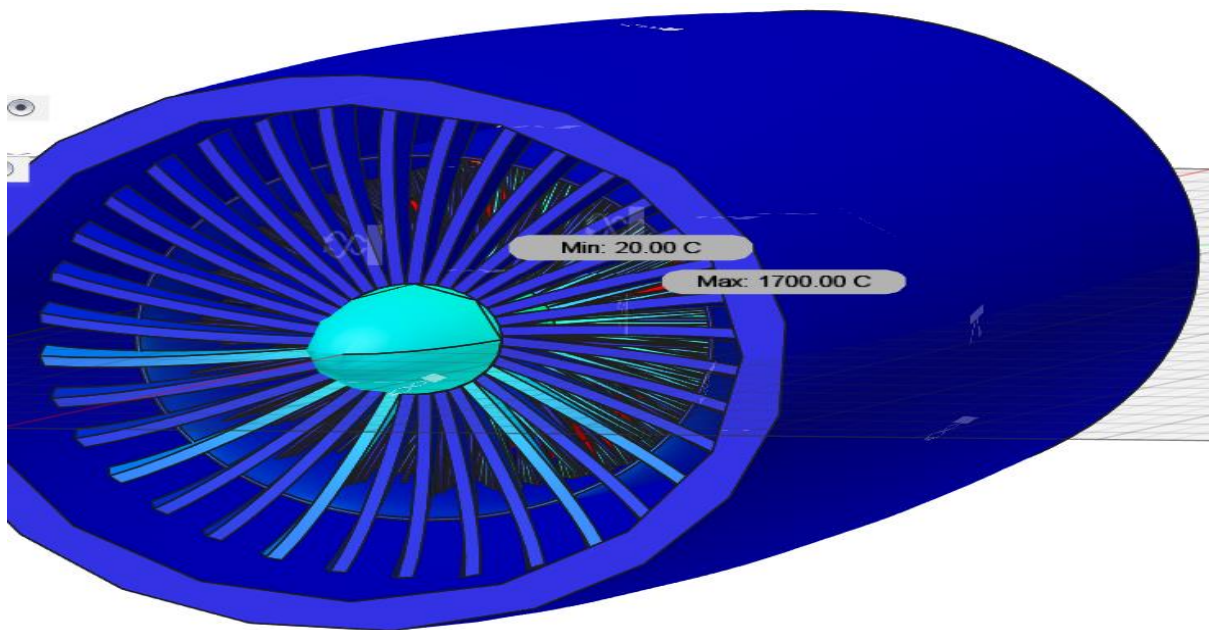
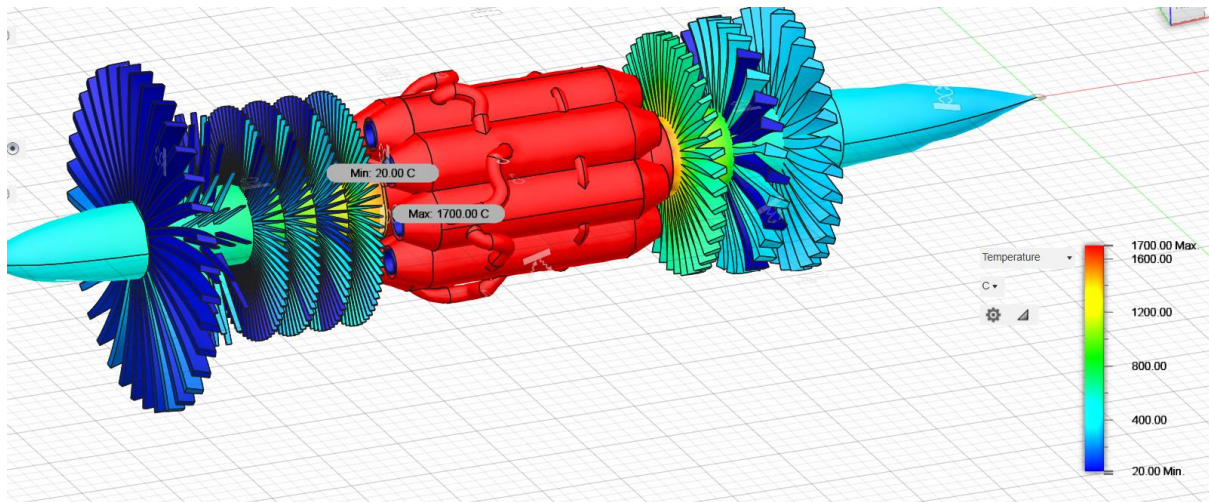
The next simulation done is a thermal analysis. Taking titanium as the material for the ramjet engine and assuming convection value to be $25\text{W/M}^2\text{K}$ (Approximate free convection value for titanium) and a Radiation emissivity value of 0.3 (Assumption taken that the titanium will not oxidize), and finally taking the internal heat of combustion to be around 2000 degrees Celsius as per usual values of combustion of jet fuel in ramjet engines. (Expertise to calculate exact values of convection value emissivity and internal heat not present so assumptions must be made). We get the following results from said simulation:

****Ramjet engine thermal analysis****



Many parts of the ramjet engine remain cool with even the front nozzle remaining at 40 degrees Celsius, and with the air speed achieved the cooling should be enough such that the titanium (1700 melting point) does not melt or stretch to too excessive of an extent. (Air flow is not considered in simulation). Along with this titanium in the simulation has been taken in direct contact with the maximum heat of 2000 degrees C while in reality a small airflow area would prevent this further reducing temperatures. The heat would also dissipate to air and to other parts of the engine. Ambient temperatures assumed are also high at 20 Degrees, while at required altitudes of operation they would likely be lower. The fuel would also additionally act as a coolant along with any propellants/water/hydraulic fluid circulating around the engine.

Similarly for a turbofan engine taking the material as aluminum (once again Al-Li simulations will not be possible), we can assume the conductivity value of free conduction to be $0.25 \text{ W/m}^2 \text{ K}$ and a radiation emissivity value 0.03. The core heat can be taken as around 1700 degrees Celsius. The parameters given show the following results:



The maximum temperature is 1700 degrees, which when combined with airflow and all previous conditions stated for ramjet will not cause liquidation of the metal (melting point of around 600 degrees) however, slight heat flux might be seen which is why joints from the engine to the wings while solid, must not be too rigid and have some amount of flex in the same logic as an aircraft has wing flex.

RESULTS

The final result of this project is 4 3d models (2 turbofan and 2 ramjet) where the ramjet engines can produce 70000N of thrust, and both engines are valid in terms of thermal integrity. Thought process has been put into design, materials, airflow, aerodynamics, features, fuel injection, simulation, etc. 3D renditions of all types of engines has also been completed. Calculations made for the project have been verified and approved. This project is ready for submission.

CONCLUSION

In conclusion, everything that has been done for the project has been meticulously detailed in this report and I do hope that it is to your satisfaction. It has brought me great joy working on this project and completing it to the best of my ability. This report is in no way to be seen as a real-life applicable engine as a lot more research beyond the scope of this internship must be done. Overall, I would like to say it has been a great honor and privilege to have been able to work with Dwello Aerospace to bring this project to life. All knowledge provided from the classes has been implemented into the engine as well as external knowledge.

For the final time this is Shlok Alva, thanking you for this opportunity.

SHLOK ALVA

REFERENCES

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NASA website – published findings and knowledge
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